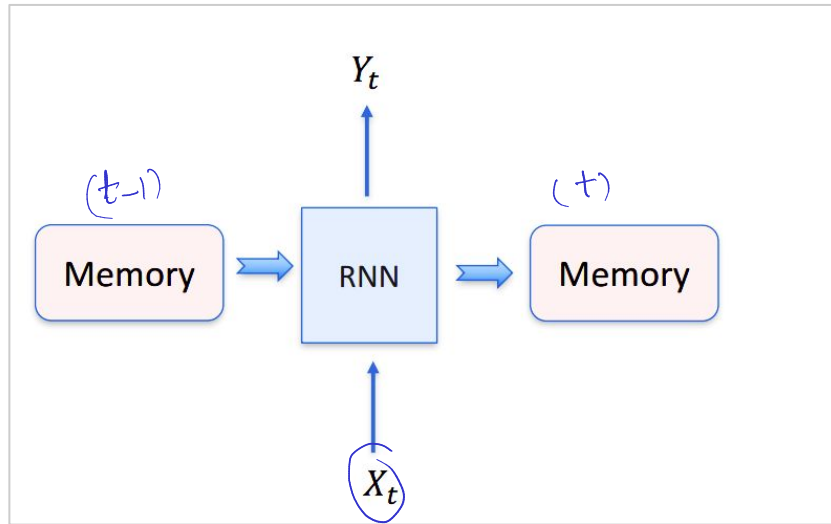


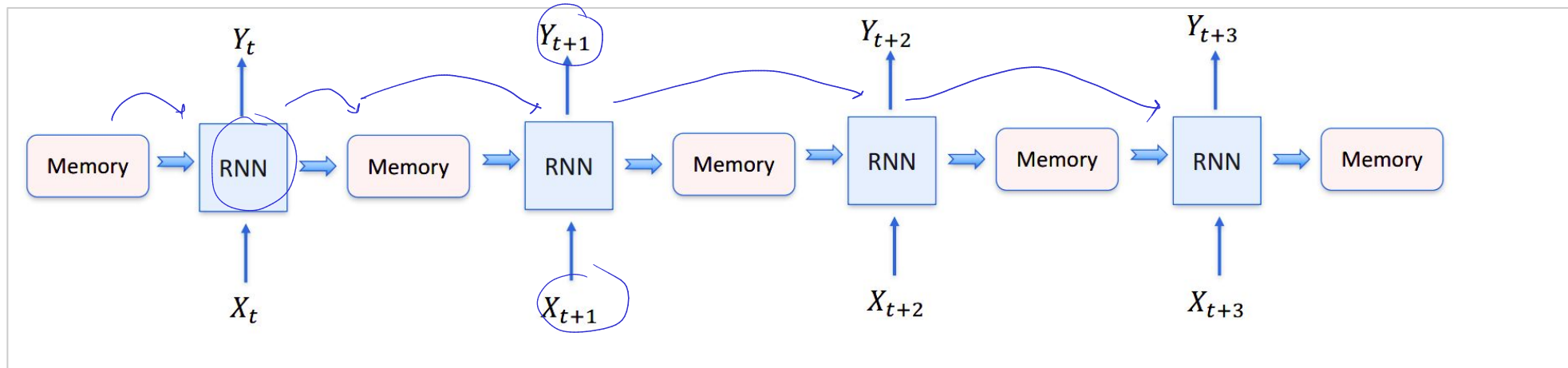
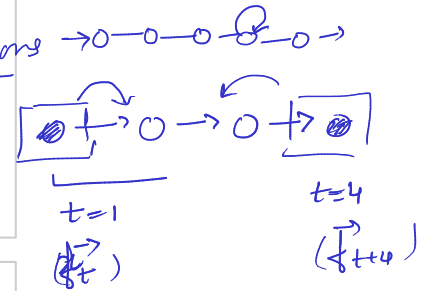
Design goals for deep neural network that processes sequential data



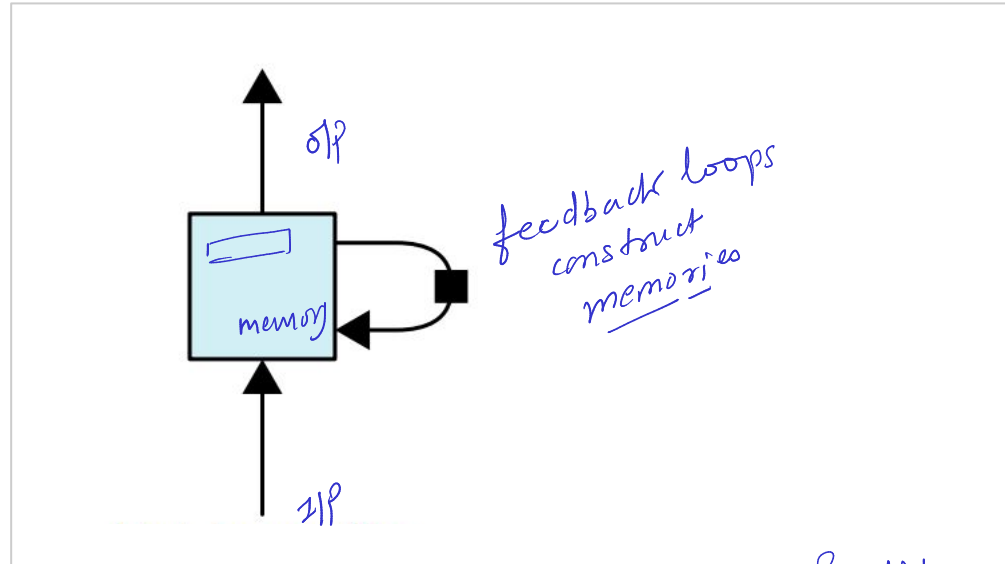
Sequential data needs a Feedback (memory) Mechanism!

our sequence model should -

- model variable length sequences
- model long term dependencies *(long) time hidden states -*
- order: connect previous information to the present task
- model sequence of events with loops *events $\Leftrightarrow$ repetitions*
 - allowing information to persist
- parameter sharing to learn shared representations

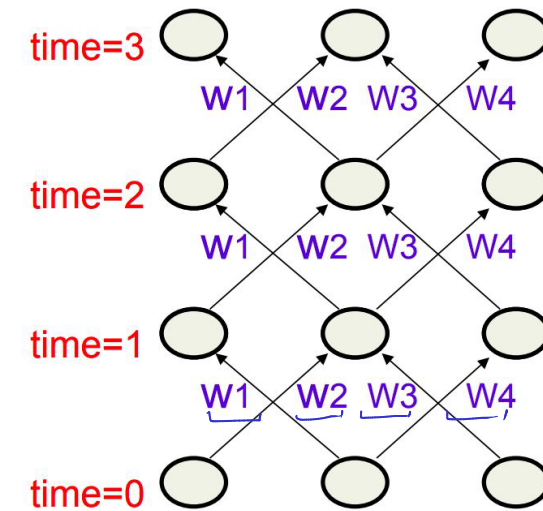
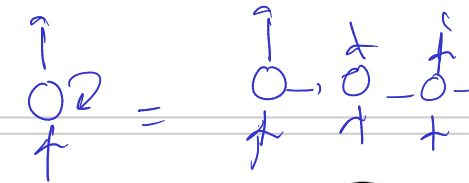
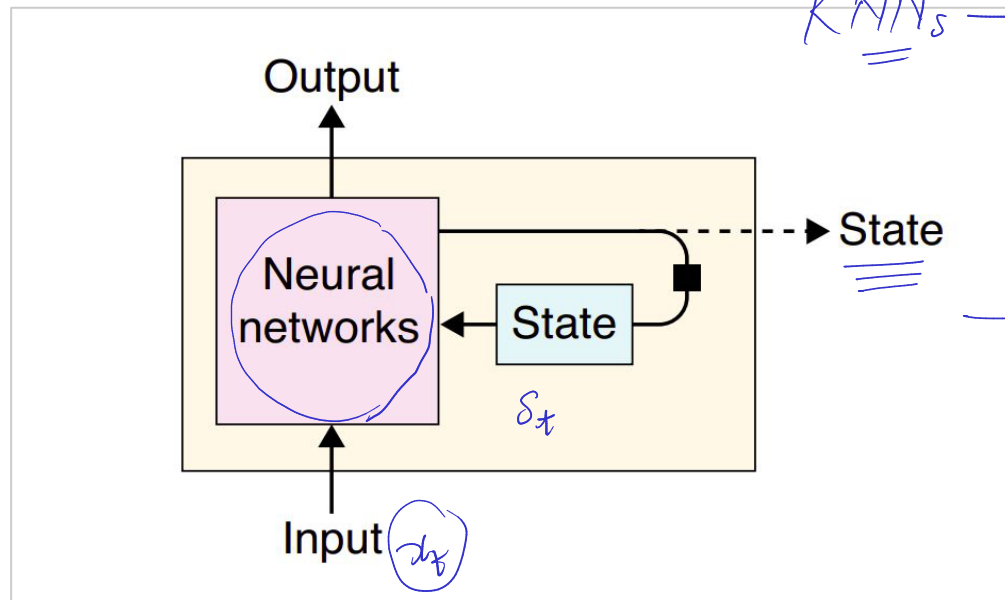


The idea of recurrence and feedback



Recurrent Neural Networks

- are deep neural networks
- contain loops for feedback mechanism
- have information retention (memory)
- can model sequences using memory



The idea of recurrence and feedback

